AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM354
Software Engineering

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 15 /Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

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Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part I – Multiple Choice Questions — 3 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 1 Mark.*

1. Which of the following characteristics of good software system indicate that errors must be minimized:
   a. Size b. Useful c. Available d. Reliable

2. Which of the below security requirements address the consistency of data
   a. Access b. Privacy c. Integrity d. Immunity

3. Which of the following terms is used to describe what symbols of a modelling language mean:
   a. Syntax b. Semantic c. Model d. Design

Part II – Short Answer Questions — 12 Marks

*Answer **all** the following short answer questions. Each question is worth 3 Marks.*

1. Based on your understanding of coupling and cohesion which of the component or service has a higher level of coupling and why?

2. What are the main drawbacks of using the Waterfall model for software development? Do you consider the Iterative and Incremental approach to be a solution? Why?

3. Explain how each of the developers, testers, and clients make use of the fit-criteria for the system requirements

4. Knowing that the customer requirements might change frequently, explain why in that case the Agile approach might be the ideal one for software development.

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Part III – Problem Solving Questions — 15 Marks

*Answer **all** the following problem-solving questions. Each question is worth 7.5 Marks.*

1. Develop a Use Case diagram for a News Reporting Company with the following requirements:
   a. The main actor is a user that can only search articles, report fake news, edit preferences.
   b. A moderator is a user that can add news, delete news, and ban reported users.

   Important: Make sure to use Generalization properly and use stereotypes at least once.

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2. Develop an activity diagram for the process of selecting the employee of the month. The diagram must include all the following:
   - Must have at least 2 roles represented by swimlanes
   - Must have at least one decision and one merge node (Example: employee might not be eligible to become employee of the month because he/she has won last month or complaints have been filed by a customer).
   - Must use the synchronization bar at least once

End of questions

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